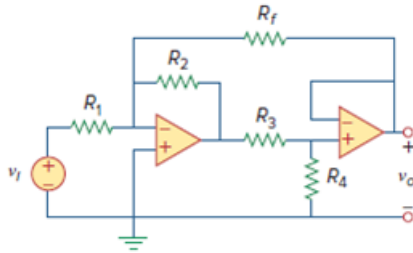


## Problem

Obtain the closed-loop voltage gain  $v_o/v_i$  of the circuit in Fig. 5.89.

**Figure 5.89:**



## Step-by-step solution

## Step 1 of 2

Let  $V_1$  Be the output of the first op-amp, and

$V_2$  Be the output of the second op-amp.

The first stage is a summer.

$$V_1 = \left( \frac{-R_2}{R_1} \right) V_i - \left( \frac{R_2}{R_f} \right) V_0 \dots\dots\dots (1)$$

The second stage is follower by voltage division.

$$V_0 = V_2 = \left( \frac{R_4}{R_3 + R_4} \right) V_1$$

$$V_1 = \left( \frac{R_3 + R_4}{R_4} \right) V_0 \dots\dots\dots (2)$$

## Step 2 of 2

From (1) and (2),

$$\left( 1 + \frac{R_3}{R_4} \right) V_0 = \left( \frac{-R_2}{R_1} \right) V_i - \left( \frac{-R_2}{R_f} \right) V_0$$

$$\left( 1 + \frac{R_3}{R_4} + \frac{R_2}{R_f} \right) V_0 = \left( \frac{-R_2}{R_1} \right) V_i$$

$$\frac{V_0}{V_i} = -\frac{R_2}{R_1} \left( \frac{1}{1 + \frac{R_3}{R_4} + \frac{R_2}{R_f}} \right)$$

Therefore,

$$\boxed{\frac{V_0}{V_i} = \frac{-R_2 R_4 R_f}{R_1 (R_2 R_4 + R_3 R_f + R_4 R_f)}}$$